WHEN LABELS LIE: UNEXPECTED OPTICAL FEATURES OF FLUORESCENT LIPID PROBES IN ELECTROFORMED MODEL MEMBRANES

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Keywords: Giant Unilamellar Vesicles (GUVs), Electroformation, Confocal Microscopy, Spectral Analysis

ABSTRACT

Giant Unilamellar Vesicles (GUVs) are versatile biomimetic systems for investigating membrane biophysics, lipid organization, phase separation, and protein interactions in a cell-like environment[1]. Among the available producing approaches, in this study we investigated the electroformation of GUVs using microfabricated interdigitated titanium/gold (Ti/Au) electrodes[2]. GUVs were prepared *via* alternating electric field hydration[1,2] from a lipid mixture composed of 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC) and 1,2-dioleoyl-sn-glycero-3-phospho ethanolamine (DOPE) doped with Atto647N probe. Intriguingly, confocal microscopy imaging and fluorimetric spectral analysis revealed a systematic, previously unreported secondary emission at shorter wavelengths alongside the expected Atto647N signal. Our data suggest that this spurious emission appears to be closely connected to the electroformation process itself, with a probable involvement of reactive oxygen species (ROS), and strictly modulated by the specific composition of the electrode material. In more detail, the phenomenon is likely driven by electrochemical processes and/or altered lipid-probe interactions within the vesicle membrane. Overall, these results emphasize the importance of rigorous spectral validation, while highlighting critical considerations for accurate multichannel imaging in model membrane systems.

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